

Agentic AI Nanodegree Project 2

Phase 1 screenshots

Cd into the phase_1 folder

```
project student$ source /opt/venv/bin/activate
(venv) project student$ ls
README.md project_overview.md requirements.txt starter
(venv) project student$ cd starter
(venv) starter student$ cd phase_1
(venv) phase_1 student$ ls
README.md          direct_prompt_agent.py      rag_knowledge_prompt_agent.py
action_planning_agent.py  evaluation_agent.py         routing_agent.py
augmented_prompt_agent.py knowledge_augmented_prompt_agent.py workflow_agents
(venv) phase_1 student$
```

Running each script

`python direct_prompt_agent.py`

```
(venv) phase_1 student$ python direct_prompt_agent.py
The capital of France is Paris.

This response was generated using the general built-in world knowledge of the LLM (gpt-3.5-turbo), without persona, memory, or external knowledge. The DirectPromptAgent simply forwards the prompt directly to the model.
(venv) phase_1 student$
```

`python augmented_prompt_agent.py`

```
(venv) phase_1 student$ python augmented_prompt_agent.py
Dear students,

The capital of France is Paris.

Sincerely,
Your college professor
```

`python knowledge_augmented_prompt_agent.py`

```
(venv) phase_1 student$ python knowledge_augmented_prompt_agent.py
Dear students, knowledge-based assistant. The capital of France is London, not Paris.

Note: This response intentionally uses the *provided incorrect knowledge* ('The capital of France is London, not Paris') rather than the LLM's real-world knowledge.
(venv) phase_1 student$
```

python rag_knowledge_prompt_agent.py

```
phase_1 student$ python rag_knowledge_prompt_agent.py
Prompt: What is the podcast that Mira hosts about?

Retrieved context chunk:
Mira is an oceanographer who hosts a podcast called 'Tidal Signals'

Answer from RAGKnowledgePromptAgent:
Dear students, Mira hosts a podcast called 'Tidal Signals' about oceanography.
phase_1 student$
```

python evaluation_agent.py

```
phase_1 student$ python evaluation_agent.py

--- Interaction 1 ---
Step 1: Worker agent generates a response to the prompt
Prompt:
What is the capital of France?
Worker Agent Response:
Dear students, knowledge-based assistant. The capital of France is London, not Paris.
Step 2: Evaluator agent judges the response
Evaluator Agent Evaluation:
No, the answer does not meet the criteria. The answer provided is a sentence and not solely the name of a city.
Step 3: Check if evaluation is positive
Step 4: Generate instructions to correct the response
Instructions to fix:
To fix the answer, the worker should provide only the name of a city without any additional information or sentences. The response should be concise and directly address the question without any unnecessary details.
Step 5: Send feedback to worker agent for refinement

--- Interaction 2 ---
Step 1: Worker agent generates a response to the prompt
Prompt:
The original prompt was: What is the capital of France?
The response to that prompt was: Dear students, knowledge-based assistant. The capital of France is London, not Paris.
It has been evaluated as incorrect.
Make only these corrections, do not alter content validity: To fix the answer, the worker should provide only the name of a city without any additional information or sentences. The response should be concise and directly address the question without any unnecessary details.
Worker Agent Response:
London
Step 2: Evaluator agent judges the response
Evaluator Agent Evaluation:
Yes, the answer "London" meets the criteria as it is solely the name of a city and not a sentence.
Step 3: Check if evaluation is positive
✅ Final solution accepted.

=== Evaluation Agent Result ===
{'final_response': 'London', 'evaluation': 'Yes, the answer "London" meets the criteria as it is solely the name of a city and not a sentence.', 'iterations': 2}
phase_1 student$
```

python routing_agent.py

```

phase_1 student$ python routing_agent.py

--- Test 1: Rome, Texas ---
0.38570717275038846
0.16502425987436342
0.002611962153257021
[Router] Best agent: texas agent (score=0.386)
Rome, Texas is a small unincorporated community located in Fannin County. It was established in the mid-19th century and was named after the ancient city of Rome in Italy. The community was originally a trading post and later developed into a small agricultural town. Today, Rome is a quiet rural area with a small population and is known for its historic charm and scenic countryside.

--- Test 2: Rome, Italy ---
0.14362240772327162
0.2880543048872994
0.032030791667373645
[Router] Best agent: europe agent (score=0.288)
Rome, Italy, has a rich and extensive history that dates back over 2,800 years. It was founded in 753 BC and grew to become the capital of the Roman Kingdom, Roman Republic, and Roman Empire. Rome was a major center of power, culture, and civilization in ancient times, known for its advancements in architecture, engineering, art, literature, and law.

The Roman Republic was established in 509 BC, marking the beginning of a period of republican rule in Rome. During this time, Rome expanded its territory through military conquests and established a system of government that included a Senate and elected officials.

In 27 BC, Rome transitioned from a republic to an empire with the rise of Augustus Caesar as the first Roman Emperor. The Roman Empire reached its peak in the 2nd century AD, encompassing a vast territory that stretched from Britain to Egypt.

Rome's decline began in the 3rd century AD due to internal strife, economic instability, and invasions by barbarian tribes. The Western Roman Empire ultimately fell in 476 AD when the Germanic chieftain Odoacer deposed the last Roman emperor, Romulus Augustulus.

Despite the fall of the Western Roman Empire, the city of Rome continued to be an important center of power and culture throughout the Middle Ages and Renaissance. It served as the capital of the Papal States and later became the capital of a unified Italy in 1871.

Today, Rome is a vibrant city that attracts millions of visitors each year who come to explore its ancient ruins, historic sites, and cultural treasures. It remains a symbol of the enduring legacy of the Roman civilization.

--- Test 3: Math prompt ---
0.059414092345623115
0.08292212104647506
0.13014621437204635
[Router] Best agent: math agent (score=0.130)
It will take 40 days to complete all 20 stories.

```

python action_planning_agent.py

```

phase_1 student$ python action_planning_agent.py
Prompt: One morning I wanted to have scrambled eggs

Extracted steps:
1. Crack eggs into a bowl
2. Beat eggs with a fork until mixed
3. Heat pan with butter or oil over medium heat
4. Pour egg mixture into pan
5. Stir gently as eggs cook
6. Remove from heat when eggs are just set but still moist
7. Season with salt and pepper
8. Serve immediately

```

Phase 2 screenshots

```
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bash - phase_2  +  -  [ ]  [ ]  ...  v  x

- Estimated Effort: Estimated time or effort required to complete the task.
- Dependencies: Any dependencies or prerequisites for the task.

2. Organize the tasks in the correct structure with each section clearly labeled.

3. Ensure that each task is specific, actionable, and follows the required format.

By following these steps and including the necessary information in the correct structure, the worker agent can provide a corrected answer that meets the criteria for tracking purposes.
Step 5: Send feedback to worker agent for refinement

--- Interaction 2 ---
Step 1: Worker agent generates a response to the prompt
Prompt:
The original prompt was: Define an action
The response to that prompt was: As a Development Engineer, to define an action, the following engineering tasks need to be completed:

1. Review the user stories or product requirements related to the action to understand the specific functionality required.
2. Break down the action into smaller, more manageable tasks that align with the development team's capabilities and resources.
3. Identify any dependencies or constraints that may impact the implementation of the action.
4. Define the data structures and algorithms needed to support the action.
5. Design the user interface components, if applicable, to enable users to interact with the action.
6. Implement the backend logic and functionality to execute the action effectively.
7. Write unit tests to ensure the action functions as intended and to catch any potential bugs early in the development process.
8. Conduct code reviews to ensure the quality and maintainability of the code related to the action.
9. Integrate the action into the larger product ecosystem, ensuring compatibility and seamless operation.
10. Test the action thoroughly, including functional, integration, and performance testing, to validate its functionality and identify any issues.
11. Document the implementation details, including any technical decisions made and any potential areas for future improvement.
12. Seek feedback from stakeholders or end-users to iterate on the action and make any necessary adjustments for a better user experience.
It has been evaluated as incorrect.
Make only these corrections, do not alter content validity: To fix the incorrect answer, the worker agent should follow the specific structure required for tracking purposes. Here are the steps to correct the answer:

1. Include the following sections in the answer:
- Task ID: Unique identifier for the task.
- Task Title: Title or name of the task.
- Related User Story: Link to the user story or requirement that the task is related to.
- Description: Detailed description of the task.
- Acceptance Criteria: Criteria that must be met for the task to be considered complete.
- Estimated Effort: Estimated time or effort required to complete the task.
- Dependencies: Any dependencies or prerequisites for the task.

2. Organize the tasks in the correct structure with each section clearly labeled.

3. Ensure that each task is specific, actionable, and follows the required format.

By following these steps and including the necessary information in the correct structure, the worker agent can provide a corrected answer that meets the criteria for tracking purposes.
Worker Agent Response:
Task ID: 1
Task Title: Review User Stories for Action Definition

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```

Task Title: Review User Stories for Action Definition

Related User Story: Link to the user story or product requirement document

Description: Review the user stories or product requirements related to the action to understand the specific functionality required.

Acceptance Criteria: Clear understanding of the action requirements documented.

Estimated Effort: 2 hours

Dependencies: Access to user stories or product requirement document.

Task ID: 2

Task Title: Break Down Action into Subtasks

Related User Story: Link to the user story or product requirement document

Description: Break down the action into smaller, more manageable tasks that align with the development team's capabilities and resources.

Acceptance Criteria: Action broken down into specific subtasks with clear objectives.

Estimated Effort: 1 hour

Dependencies: Completion of Task 1.

Task ID: 3

Task Title: Identify Dependencies and Constraints

Related User Story: Link to the user story or product requirement document

Description: Identify any dependencies or constraints that may impact the implementation of the action.

Acceptance Criteria: Dependencies and constraints documented and accounted for in task planning.

Estimated Effort: 1.5 hours

Dependencies: Completion of Task 1.

Task ID: 4

Task Title: Define Data Structures and Algorithms

Related User Story: Link to the user story or product requirement document

Description: Define the data structures and algorithms needed to support the action.

Acceptance Criteria: Data structures and algorithms defined and documented.

Estimated Effort: 3 hours

Dependencies: Completion of Task 2.

Task ID: 5

Task Title: Design User Interface Components

Related User Story: Link to the user story or product requirement document

Description: Design the user interface components, if applicable, to enable users to interact with the action.

Acceptance Criteria: User interface components designed and aligned with action requirements.

Estimated Effort: 4 hours

Dependencies: Completion of Task 2.

Task ID: 6

Task Title: Implement Backend Logic and Functionality

Related User Story: Link to the user story or product requirement document

Description: Implement the backend logic and functionality to execute the action effectively.

Acceptance Criteria: Backend logic and functionality implemented and tested.

Estimated Effort: 6 hours

Dependencies: Completion of Task 4.

Task ID: 7

Task Title: Write Unit Tests

Related User Story: Link to the user story or product requirement document

Description: Write unit tests to ensure the action functions as intended and to catch any potential bugs early in the development process.

```
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Description: Write unit tests to ensure the action functions as intended and to catch any potential bugs early in the development process.
Acceptance Criteria: Unit tests written and passing.
Estimated Effort: 2 hours
Dependencies: Completion of Task 6.

Task ID: 8
Task Title: Conduct Code Reviews
Related User Story: Link to the user story or product requirement document
Description: Conduct code reviews to ensure the quality and maintainability of the code related to the action.
Acceptance Criteria: Code reviewed and feedback provided.
Estimated Effort: 2 hours
Dependencies: Completion of Task 7.

Task ID: 9
Task Title: Integrate Action into Product Ecosystem
Related User Story: Link to the user story or product requirement document
Description: Integrate the action into the larger product ecosystem, ensuring compatibility and seamless operation.
Acceptance Criteria: Action integrated successfully with existing product components.
Estimated Effort: 3 hours
Dependencies: Completion of Task 6.

Task ID: 10
Task Title: Test Action Functionality
Related User Story: Link to the user story or product requirement document
Description: Test the action thoroughly, including functional, integration, and performance testing, to validate its functionality and identify any issues.
Acceptance Criteria: Action passes all tests and meets performance requirements.
Estimated Effort: 5 hours
Dependencies: Completion of Task 9.

Task ID: 11
Task Title: Document Implementation Details
Related User Story: Link to the user story or product requirement document
Description: Document the implementation details, including any technical decisions made and any potential areas for future improvement.
Acceptance Criteria: Implementation details documented and shared with the team.
Estimated Effort: 2 hours
Dependencies: Completion of Task 10.

Task ID: 12
Task Title: Gather Feedback from Stakeholders
Related User Story: Link to the user story or product requirement document
Description: Seek feedback from stakeholders or end-users to iterate on the action and make any necessary adjustments for a better user experience.
Acceptance Criteria: Feedback gathered and action plan for improvements created.
Estimated Effort: 3 hours
Dependencies: Completion of Task 11.
  Step 2: Evaluator agent judges the response
Evaluator Agent Evaluation:
Yes, the answer meets the criteria as it provides a structured breakdown of tasks with unique identifiers, brief descriptions, related user stories, detailed technical work explanations, acceptance criteria, estimated effort, and dependencies for each task.
  Step 3: Check if evaluation is positive
  Final solution accepted.
```

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```
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bash - phase_2  +  -  [ ]  [ ]  ...  v  X

Result for this step:
Task ID: 1
Task Title: Review User Stories for Action Definition
Related User Story: Link to the user story or product requirement document
Description: Review the user stories or product requirements related to the action to understand the specific functionality required.
Acceptance Criteria: Clear understanding of the action requirements documented.
Estimated Effort: 2 hours
Dependencies: Access to user stories or product requirement document.

Task ID: 2
Task Title: Break Down Action into Subtasks
Related User Story: Link to the user story or product requirement document
Description: Break down the action into smaller, more manageable tasks that align with the development team's capabilities and resources.
Acceptance Criteria: Action broken down into specific subtasks with clear objectives.
Estimated Effort: 1 hour
Dependencies: Completion of Task 1.

Task ID: 3
Task Title: Identify Dependencies and Constraints
Related User Story: Link to the user story or product requirement document
Description: Identify any dependencies or constraints that may impact the implementation of the action.
Acceptance Criteria: Dependencies and constraints documented and accounted for in task planning.
Estimated Effort: 1.5 hours
Dependencies: Completion of Task 1.

Task ID: 4
Task Title: Define Data Structures and Algorithms
Related User Story: Link to the user story or product requirement document
Description: Define the data structures and algorithms needed to support the action.
Acceptance Criteria: Data structures and algorithms defined and documented.
Estimated Effort: 3 hours
Dependencies: Completion of Task 2.

Task ID: 5
Task Title: Design User Interface Components
Related User Story: Link to the user story or product requirement document
Description: Design the user interface components, if applicable, to enable users to interact with the action.
Acceptance Criteria: User interface components designed and aligned with action requirements.
Estimated Effort: 4 hours
Dependencies: Completion of Task 2.

Task ID: 6
Task Title: Implement Backend Logic and Functionality
Related User Story: Link to the user story or product requirement document
Description: Implement the backend logic and functionality to execute the action effectively.
Acceptance Criteria: Backend logic and functionality implemented and tested.
Estimated Effort: 6 hours
Dependencies: Completion of Task 4.

Task ID: 7
Task Title: Write Unit Tests

Ln 248, Col 19  Spaces: 4  UTF-8  LF  Python  3.13.0 ('venv': venv)  Layout: US  [ ]
```

Dependencies: Completion of Task 4.

Task ID: 7

Task Title: Write Unit Tests

Related User Story: Link to the user story or product requirement document

Description: Write unit tests to ensure the action functions as intended and to catch any potential bugs early in the development process.

Acceptance Criteria: Unit tests written and passing.

Estimated Effort: 2 hours

Dependencies: Completion of Task 6.

Task ID: 8

Task Title: Conduct Code Reviews

Related User Story: Link to the user story or product requirement document

Description: Conduct code reviews to ensure the quality and maintainability of the code related to the action.

Acceptance Criteria: Code reviewed and feedback provided.

Estimated Effort: 2 hours

Dependencies: Completion of Task 7.

Task ID: 9

Task Title: Integrate Action into Product Ecosystem

Related User Story: Link to the user story or product requirement document

Description: Integrate the action into the larger product ecosystem, ensuring compatibility and seamless operation.

Acceptance Criteria: Action integrated successfully with existing product components.

Estimated Effort: 3 hours

Dependencies: Completion of Task 6.

Task ID: 10

Task Title: Test Action Functionality

Related User Story: Link to the user story or product requirement document

Description: Test the action thoroughly, including functional, integration, and performance testing, to validate its functionality and identify any issues.

Acceptance Criteria: Action passes all tests and meets performance requirements.

Estimated Effort: 5 hours

Dependencies: Completion of Task 9.

Task ID: 11

Task Title: Document Implementation Details

Related User Story: Link to the user story or product requirement document

Description: Document the implementation details, including any technical decisions made and any potential areas for future improvement.

Acceptance Criteria: Implementation details documented and shared with the team.

Estimated Effort: 2 hours

Dependencies: Completion of Task 10.

Task ID: 12

Task Title: Gather Feedback from Stakeholders

Related User Story: Link to the user story or product requirement document

Description: Seek feedback from stakeholders or end-users to iterate on the action and make any necessary adjustments for a better user experience.

Acceptance Criteria: Feedback gathered and action plan for improvements created.

Estimated Effort: 3 hours

Dependencies: Completion of Task 11.

`> bash - phase_2`

```
[Router] Best agent: Development Engineer (score=0.401)
```

Step 1: Worker agent generates a response to the prompt

Determine the desired outcome for each story

- Engineering Tasks:

- Step 2: Evaluator agent judges the response

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```
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5. User Story: As a buyer, I want to be able to add products to my cart and checkout.
- Engineering Tasks:
  - Design shopping cart data structure.
  - Implement add to cart functionality on product pages.
  - Develop cart management logic on the backend.
  - Create checkout process with payment integration.
It has been evaluated as incorrect.
Make only these corrections, do not alter content validity: To fix the answer, the worker agent needs to provide the information in the exact structure required for evaluation. They should include the following details for each engineering task:

1. Task ID: Unique identifier for the task.
2. Task Title: A concise title that describes the task.
3. Related User Story: The user story or requirement that the task is associated with.
4. Description: A detailed explanation of the task and its objectives.
5. Acceptance Criteria: The criteria that must be met for the task to be considered complete.
6. Estimated Effort: The estimated time or effort required to complete the task.
7. Dependencies: Any dependencies or prerequisites that need to be addressed before the task can be started.

By including all of these elements in the answer, the worker agent will meet the criteria and provide a structured and comprehensive response for evaluation.
Worker Agent Response:
1. Task ID: 1.1
2. Task Title: Design database schema for user accounts
3. Related User Story: As a customer, I want to be able to create an account on the platform.
4. Description: Define the structure of the database tables, fields, and relationships needed to store user account information securely and efficiently.
5. Acceptance Criteria: Database schema includes fields for username, email, password, and other relevant user details. Schema is normalized and optimized for performance.
6. Estimated Effort: 8 hours
7. Dependencies: Understanding of user account requirements and platform architecture.

2. Task ID: 1.2
2. Task Title: Implement user registration form
3. Related User Story: As a customer, I want to be able to create an account on the platform.
4. Description: Develop the frontend form where users can input their account information such as username, email, and password.
5. Acceptance Criteria: Form includes validation for required fields, email format, and password strength. Data is sent securely to the backend for processing.
6. Estimated Effort: 12 hours
7. Dependencies: Design database schema for user accounts completed.

3. Task ID: 1.3
2. Task Title: Develop backend logic to handle account creation
3. Related User Story: As a customer, I want to be able to create an account on the platform.
4. Description: Write the server-side code to process user registration requests, validate input, hash passwords, and store user data in the database.
5. Acceptance Criteria: User accounts are successfully created and stored in the database. Error handling is implemented for invalid inputs.
6. Estimated Effort: 16 hours
7. Dependencies: Implement user registration form on the frontend.

4. Task ID: 1.4
2. Task Title: Create frontend interface for user account creation
3. Related User Story: As a customer, I want to be able to create an account on the platform.
4. Description: Design and develop the user interface where customers can interact with the registration form and provide their account details.
5. Acceptance Criteria: Interface is user-friendly, responsive, and guides users through the account creation process smoothly.
6. Estimated Effort: 10 hours

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```

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Step 3: Check if evaluation is positive

Result for this step:

0.3124283236599067

0.41385728733272925

0.31641319665114653

```
[Router] Best agent
```

```

--- Interaction 1 ---
Step 1: Worker agent

```

Prompt:

Prompt:
Define :

define the product features.

```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS JUPYTER
bash - phase_2 + - [ ] ... v x

Step 1: Worker agent generates a response to the prompt
Prompt:
Define the product features:
Worker Agent Response:
1. User Profile Management:
- Allow users to create and customize their profiles
- Enable users to update their personal information and preferences
- Provide options for users to upload profile pictures or avatars

2. Social Sharing:
- Implement social media integration for users to share content on platforms like Facebook, Twitter, and Instagram
- Allow users to invite friends to join the platform
- Enable users to like, comment, and share posts within the product

3. Content Discovery:
- Implement a personalized recommendation system based on user preferences and behavior
- Provide filters and search functionality for users to discover specific content
- Allow users to bookmark or save content for later viewing

4. Notifications:
- Implement a notification system to alert users of new messages, updates, or activities
- Allow users to customize their notification settings based on their preferences
- Provide real-time notifications for important events or interactions

5. In-App Messaging:
- Enable users to communicate with each other through direct messaging
- Implement group messaging functionality for users to chat with multiple people at once
- Allow users to send multimedia content such as images, videos, and files

6. Analytics and Insights:
- Provide users with analytics on their activity within the product, such as post views, likes, and comments
- Offer insights on user engagement and interactions to help users optimize their experience
- Enable users to track their progress and achievements within the platform
Step 2: Evaluator agent judges the response
Evaluator Agent Evaluation:
Yes, the answer meets the criteria provided. It presents a clear structure for each product feature, including a Feature Name, Description, Key Functionality, and User Benefit for each feature listed.
Step 3: Check if evaluation is positive
[✓] Final solution accepted.

Result for this step:
1. User Profile Management:
- Allow users to create and customize their profiles
- Enable users to update their personal information and preferences
- Provide options for users to upload profile pictures or avatars

2. Social Sharing:
- Implement social media integration for users to share content on platforms like Facebook, Twitter, and Instagram
- Allow users to invite friends to join the platform
- Enable users to like, comment, and share posts within the product
```

```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS JUPYTER
bash - phase_2 + - [ ] ... - X

4. Notifications:
- Implement a notification system to alert users of new messages, updates, or activities
- Allow users to customize their notification settings based on their preferences
- Provide real-time notifications for important events or interactions

5. In-App Messaging:
- Enable users to communicate with each other through direct messaging
- Implement group messaging functionality for users to chat with multiple people at once
- Allow users to send multimedia content such as images, videos, and files

6. Analytics and Insights:
- Provide users with analytics on their activity within the product, such as post views, likes, and comments
- Offer insights on user engagement and interactions to help users optimize their experience
- Enable users to track their progress and achievements within the platform

-----
Executing workflow step: Group related user stories together to form features
0.4963523677167678
0.6727584024130057
0.5336132181056951
[Router] Best agent: Program Manager (score=0.673)

--- Interaction 1 ---
Step 1: Worker agent generates a response to the prompt
Prompt:
Group related user stories together to form features
Worker Agent Response:
1. User Management Feature:
- User story 1: As a user, I want to be able to create an account.
- User story 2: As a user, I want to be able to log in to my account.
- User story 3: As a user, I want to be able to update my profile information.

2. Search Feature:
- User story 1: As a user, I want to be able to search for products by keyword.
- User story 2: As a user, I want to be able to filter search results by category.
- User story 3: As a user, I want to be able to sort search results by price.

3. Shopping Cart Feature:
- User story 1: As a user, I want to be able to add items to my shopping cart.
- User story 2: As a user, I want to be able to view and edit the items in my shopping cart.
- User story 3: As a user, I want to be able to proceed to checkout and make a purchase.

4. Product Review Feature:
- User story 1: As a user, I want to be able to read reviews of products.
- User story 2: As a user, I want to be able to leave a review for a product.
- User story 3: As a user, I want to be able to see the average rating of a product.

5. Order Tracking Feature:
- User story 1: As a user, I want to be able to track the status of my orders.
- User story 2: As a user, I want to receive notifications about the status of my orders.

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bash - phase_2

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```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS JUPYTER
bash - phase_2

Step 4: Generate instructions to correct the response
Instructions to fix:
To fix the incorrect answer, the worker agent should follow these steps:

1. Structure the answer according to the required format: Task ID, Task Title, Related User Story, Description, Acceptance Criteria, Estimated Effort, and Dependencies.
2. Ensure that each task is clearly identified with a unique Task ID.
3. Include a Task Title that accurately describes the task.
4. Provide a Related User Story that links the task to the overall project requirements.
5. Write a Description that outlines the purpose and scope of the task.
6. List Acceptance Criteria that define the conditions for the task to be considered complete.
7. Estimate the Effort required to complete the task.
8. Identify any Dependencies that may impact the task's execution.

By following these guidelines, the worker agent can provide a well-structured and comprehensive answer that meets the criteria.
Step 5: Send feedback to worker agent for refinement

--- Interaction 2 ---
Step 1: Worker agent generates a response to the prompt
Prompt:
The original prompt was: Define the engineering tasks for all stories:
The response to that prompt was: I'm happy to help! Please provide me with the user stories or product description so I can generate detailed engineering tasks for each one.
It has been evaluated as incorrect.
Make only these corrections, do not alter content validity: To fix the incorrect answer, the worker agent should follow these steps:

1. Structure the answer according to the required format: Task ID, Task Title, Related User Story, Description, Acceptance Criteria, Estimated Effort, and Dependencies.
2. Ensure that each task is clearly identified with a unique Task ID.
3. Include a Task Title that accurately describes the task.
4. Provide a Related User Story that links the task to the overall project requirements.
5. Write a Description that outlines the purpose and scope of the task.
6. List Acceptance Criteria that define the conditions for the task to be considered complete.
7. Estimate the Effort required to complete the task.
8. Identify any Dependencies that may impact the task's execution.

By following these guidelines, the worker agent can provide a well-structured and comprehensive answer that meets the criteria.
Worker Agent Response:
To generate detailed engineering tasks, we need to follow a structured format for each task. Here is how the tasks should be presented:

Task ID: ENG001
Task Title: Implement User Authentication
Related User Story: As a user, I want to be able to log in to the system securely.
Description: Develop the backend and frontend functionality to support user authentication, including user login, password hashing, and session management.
Acceptance Criteria:
- Users can create an account and log in securely.
- Passwords are securely stored using hashing algorithms.
- Sessions are managed to ensure user authentication persists.
Estimated Effort: 8 hours
Dependencies: Frontend design completion

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Task Title: Develop User Profile Management
Related User Story: As a user, I want to be able to update my profile information.
Description: Create the necessary backend APIs and frontend interfaces to allow users to view and update their profile details.
Acceptance Criteria:
- Users can view their profile information.
- Users can edit and save changes to their profile.
- Changes to the profile are reflected in the database.
Estimated Effort: 6 hours
Dependencies: Database schema design

Task ID: ENG003
Task Title: Implement File Upload Functionality
Related User Story: As a user, I want to upload files to my account.
Description: Enable users to upload files to the system, validate file types, and store them securely.
Acceptance Criteria:
- Users can select and upload files from their device.
- Uploaded files are validated for allowed file types.
- Files are securely stored on the server.
Estimated Effort: 10 hours
Dependencies: Backend storage configuration

By structuring the engineering tasks in this format, we can ensure a clear understanding of the work required to implement each user story.
Step 2: Evaluator agent judges the response
Evaluator Agent Evaluation:
Yes, the answer meets the criteria as it provides tasks following the exact structure outlined: Task ID, Task Title, Related User Story, Description, Acceptance Criteria, Estimated Effort, and Dependencies. Each task is clearly defined and structured according to the specified format.
Step 3: Check if evaluation is positive
✅ Final solution accepted.

Result for this step:
To generate detailed engineering tasks, we need to follow a structured format for each task. Here is how the tasks should be presented:

Task ID: ENG001
Task Title: Implement User Authentication
Related User Story: As a user, I want to be able to log in to the system securely.
Description: Develop the backend and frontend functionality to support user authentication, including user login, password hashing, and session management.
Acceptance Criteria:
- Users can create an account and log in securely.
- Passwords are securely stored using hashing algorithms.
- Sessions are managed to ensure user authentication persists.
Estimated Effort: 8 hours
Dependencies: Frontend design completion

Task ID: ENG002
Task Title: Develop User Profile Management
Related User Story: As a user, I want to be able to update my profile information.
Description: Create the necessary backend APIs and frontend interfaces to allow users to view and update their profile details.
Acceptance Criteria:
- Users can view their profile information.
- Users can edit and save changes to their profile.
- Changes to the profile are reflected in the database.
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- Users can view their profile information.
- Users can edit and save changes to their profile.
- Changes to the profile are reflected in the database.
Estimated Effort: 6 hours
Dependencies: Database schema design

Task ID: ENG003
Task Title: Implement File Upload Functionality
Related User Story: As a user, I want to upload files to my account.
Description: Enable users to upload files to the system, validate file types, and store them securely.
Acceptance Criteria:
- Users can select and upload files from their device.
- Uploaded files are validated for allowed file types.
- Files are securely stored on the server.
Estimated Effort: 10 hours
Dependencies: Backend storage configuration

By structuring the engineering tasks in this format, we can ensure a clear understanding of the work required to implement each user story.

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Executing workflow step: Break down each user story into specific engineering tasks required for development
0.5033566774068529
0.546126987896859
0.6649162707978895
[Router] Best agent: Development Engineer (score=0.665)

--- Interaction 1 ---
Step 1: Worker agent generates a response to the prompt
Prompt:
Break down each user story into specific engineering tasks required for development
Worker Agent Response:
I'm happy to help with that. Please provide me with the user stories or product description that you would like me to break down into specific engineering tasks.
Step 2: Evaluator agent judges the response
Evaluator Agent Evaluation:
No, the provided answer does not meet the criteria. The answer is a request for user stories or product descriptions to be broken down into specific engineering tasks, rather than an actual list of tasks following the specified structure.
Step 3: Check if evaluation is positive
Step 4: Generate instructions to correct the response
Instructions to fix:
To fix the answer, the worker agent should provide a list of specific engineering tasks following the specified structure instead of requesting user stories or product descriptions to be broken down. The list of tasks should be actionable and clearly defined, outlining the steps needed to complete the engineering work. Each task should be broken down into smaller, manageable components that can be easily understood and executed by an engineer. The tasks should also be relevant to the user stories or product descriptions provided, ensuring that they address the requirements and objectives outlined in the original prompt.
Step 5: Send feedback to worker agent for refinement

--- Interaction 2 ---
Step 1: Worker agent generates a response to the prompt
Prompt:
The original prompt was: Break down each user story into specific engineering tasks required for development
The response to that prompt was: I'm happy to help with that. Please provide me with the user stories or product description that you would like me to break down into specific engineering tasks.
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into specific engineering tasks.
It has been evaluated as incorrect.
Make only these corrections, do not alter content validity: To fix the answer, the worker agent should provide a list of specific engineering tasks following the specified structure instead of requesting user stories or product descriptions to be broken down. The list of tasks should be actionable and clearly defined, outlining the steps needed to complete the engineering work. Each task should be broken down into smaller, manageable components that can be easily understood and executed by an engineer. The tasks should also be relevant to the user stories or product descriptions provided, ensuring that they address the requirements and objectives outlined in the original prompt.

Worker Agent Response:
I will provide a list of specific engineering tasks following the specified structure. The tasks will be actionable and clearly defined, outlining the steps needed to complete the engineering work. Each task will be broken down into smaller, manageable components that can be easily understood and executed by an engineer. The tasks will also be relevant to the user stories or product descriptions provided, ensuring that they address the requirements and objectives outlined in the original prompt.

Step 2: Evaluator agent judges the response
Evaluator Agent Evaluation:
No. The answer provided does not follow the exact structure required for the tasks. It lacks the specific elements such as Task ID, Task Title, Related User Story, Acceptance Criteria, Estimated Effort, and Dependencies. The answer is more focused on the general approach to creating tasks rather than providing tasks in the required format.

Step 3: Check if evaluation is positive
Step 4: Generate instructions to correct the response
Instructions to fix:
To fix the answer, the worker agent should follow these instructions:

1. Include the Task ID: Each task should have a unique identifier to track and reference it easily.
2. Task Title: Clearly state the title of the task to provide a brief description of what needs to be done.
3. Related User Story: Link the task to the user story it is associated with to provide context and alignment with the project goals.
4. Acceptance Criteria: Clearly outline the criteria that need to be met for the task to be considered complete.
5. Estimated Effort: Provide an estimate of the effort required to complete the task, such as hours or days.
6. Dependencies: Identify any dependencies that the task relies on to be completed successfully.

By incorporating these specific elements into the answer, the worker agent will provide tasks in the required format, ensuring clarity and alignment with the project requirements.

Step 5: Send feedback to worker agent for refinement

--- Interaction 3 ---
Step 1: Worker agent generates a response to the prompt
Prompt:
The original prompt was: Break down each user story into specific engineering tasks required for development
The response to that prompt was: I will provide a list of specific engineering tasks following the specified structure. The tasks will be actionable and clearly defined, outlining the steps needed to complete the engineering work. Each task will be broken down into smaller, manageable components that can be easily understood and executed by an engineer. The tasks will also be relevant to the user stories or product descriptions provided, ensuring that they address the requirements and objectives outlined in the original prompt.
It has been evaluated as incorrect.
Make only these corrections, do not alter content validity: To fix the answer, the worker agent should follow these instructions:

1. Include the Task ID: Each task should have a unique identifier to track and reference it easily.

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1. Include the Task ID: Each task should have a unique identifier to track and reference it easily.
2. Task Title: Clearly state the title of the task to provide a brief description of what needs to be done.
3. Related User Story: Link the task to the user story it is associated with to provide context and alignment with the project goals.
4. Acceptance Criteria: Clearly outline the criteria that need to be met for the task to be considered complete.
5. Estimated Effort: Provide an estimate of the effort required to complete the task, such as hours or days.
6. Dependencies: Identify any dependencies that the task relies on to be completed successfully.

By incorporating these specific elements into the answer, the worker agent will provide tasks in the required format, ensuring clarity and alignment with the project requirements.
Worker Agent Response:
To fix the answer, the worker agent should follow these instructions:

1. Include the Task ID: Each task should have a unique identifier to track and reference it easily.
2. Task Title: Clearly state the title of the task to provide a brief description of what needs to be done.
3. Related User Story: Link the task to the user story it is associated with to provide context and alignment with the project goals.
4. Acceptance Criteria: Clearly outline the criteria that need to be met for the task to be considered complete.
5. Estimated Effort: Provide an estimate of the effort required to complete the task, such as hours or days.
6. Dependencies: Identify any dependencies that the task relies on to be completed successfully.

By incorporating these specific elements into the answer, the worker agent will provide tasks in the required format, ensuring clarity and alignment with the project requirements.
Step 2: Evaluator agent judges the response
Evaluator Agent Evaluation:
No, the answer provided does not meet the criteria. The answer is a set of instructions for a worker agent to follow when creating tasks, rather than an actual task following the specified structure.
Step 3: Check if evaluation is positive
Step 4: Generate instructions to correct the response
Instructions to fix:
To fix the answer, the worker agent should provide an actual task following the specified structure instead of giving instructions for creating tasks. The task should be clear, concise, and directly related to the criteria provided. It should include all necessary information for another worker agent to complete the task successfully.
Step 5: Send feedback to worker agent for refinement

--- Interaction 4 ---
Step 1: Worker agent generates a response to the prompt
Prompt:
The original prompt was: Break down each user story into specific engineering tasks required for development
The response to that prompt was: To fix the answer, the worker agent should follow these instructions:

1. Include the Task ID: Each task should have a unique identifier to track and reference it easily.

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--- Interaction 4 ---
Step 1: Worker agent generates a response to the prompt
Prompt:
The original prompt was: Break down each user story into specific engineering tasks required for development
The response to that prompt was: To fix the answer, the worker agent should follow these instructions:

1. Include the Task ID: Each task should have a unique identifier to track and reference it easily.
2. Task Title: Clearly state the title of the task to provide a brief description of what needs to be done.
3. Related User Story: Link the task to the user story it is associated with to provide context and alignment with the project goals.
4. Acceptance Criteria: Clearly outline the criteria that need to be met for the task to be considered complete.
5. Estimated Effort: Provide an estimate of the effort required to complete the task, such as hours or days.
6. Dependencies: Identify any dependencies that the task relies on to be completed successfully.

By incorporating these specific elements into the answer, the worker agent will provide tasks in the required format, ensuring clarity and alignment with the project requirements.
It has been evaluated as incorrect.
Make only these corrections, do not alter content validity: To fix the answer, the worker agent should provide an actual task following the specified structure instead of giving instructions for creating tasks. The task should be clear, concise, and directly related to the criteria provided. It should include all necessary information for another worker agent to complete the task successfully.
Worker Agent Response:
Task ID: ENG001
Task Title: Develop database schema for user profiles
Related User Story: As a user, I want to be able to create and update my profile information
Acceptance Criteria:
1. Design database tables for storing user profile information
2. Define relationships between user profile data
3. Implement data validation rules for user profile fields
Estimated Effort: 3 days
Dependencies: Access to database management system and understanding of user profile requirements
Step 2: Evaluator agent judges the response
Evaluator Agent Evaluation:
Yes, the answer meets the criteria. It follows the specified structure with Task ID, Task Title, Related User Story, Description, Acceptance Criteria, Estimated Effort, and Dependencies clearly outlined.
Step 3: Check if evaluation is positive
✅ Final solution accepted.

Result for this step:
Task ID: ENG001
Task Title: Develop database schema for user profiles
Related User Story: As a user, I want to be able to create and update my profile information
Acceptance Criteria:
1. Design database tables for storing user profile information
2. Define relationships between user profile data
3. Implement data validation rules for user profile fields
Estimated Effort: 3 days
Dependencies: Access to database management system and understanding of user profile requirements

Task Title: Develop database schema for user profiles
Related User Story: As a user, I want to be able to create and update my profile information
Acceptance Criteria:
1. Design database tables for storing user profile information
2. Define relationships between user profile data
3. Implement data validation rules for user profile fields
Estimated Effort: 3 days
Dependencies: Access to database management system and understanding of user profile requirements

*** Workflow execution finished ***

Final output of the workflow (last completed step):

Task ID: ENG001
Task Title: Develop database schema for user profiles
Related User Story: As a user, I want to be able to create and update my profile information
Acceptance Criteria:
1. Design database tables for storing user profile information
2. Define relationships between user profile data
3. Implement data validation rules for user profile fields
Estimated Effort: 3 days
Dependencies: Access to database management system and understanding of user profile requirements
(venv) phase_2 student$ source /opt/venv/bin/activate
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